

Crystalline coordination framework endowed with dynamic gate-opening behaviour by being downsized to a thin film

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1. Materials and Methods

General. All the reagents and solvents were purchased from Wako Pure Chemical Industries Ltd., TCI Co., Ltd., Alfa Aesar Co, and Sigma-Aldrich Chemical Co. and used without further purification. The substrates were prepared by evaporating a 5-nm chromium buffer followed by 100 nm of gold onto polished silicon wafers (Au/Cr/Si, purchased from Nilaco Corp.). The typical substrate size was $20 \times 15 \times 0.5 \text{ mm}^3$. H_2 -annealing was performed on the substrate before the thin-film preparation. Infrared reflection absorption spectra (IRRAS) were collected with a Thermo Nicolet NEXUS 670 FT-IR spectrometer and Thermo Scientific Spectra-Tech Collector diffuse reflectance accessory (incident angle = 80°). Raman spectra were recorded with a JASCO NRS-1000 using a microscope. A CVI Melles Griot 25 LHP 925 He–Ne laser provided the exciting line (632.8 nm). Thermogravimetric analysis was performed with a TG-DTA 2000SA (NETZSCH Japan) at a heating rate of 5 K per min in a constant flow of N_2 . X-ray photoelectron spectroscopy (XPS) measurements were performed using an ESCA-3400 spectrometer (Shimadzu Corp.) equipped with Mg $K\alpha$ radiation as the exciting source (1253.6 eV). The binding energies were calibrated with the Au $4f_{7/2}$ signal (84.0 eV). Atomic force microscopy (AFM) images were obtained in tapping mode using NanoScope IIIa equipment (Bruker) with a Si cantilever. Adsorption isotherms for ethanol (EtOH), acetonitrile (MeCN) and benzene of the thin film sample were obtained with a BELQCM apparatus (MicrotracBEL Corp.). A sample fabricated on an Au electrode (5 mm ϕ) on a quartz tip (14 mm ϕ) was used. The vapour pressure in He gas was controlled by mass flow controller (MFC), bubbler, and condenser installed on a BEL-Flow apparatus (MicrotracBEL Corp.).

Single-crystal X-ray diffraction (XRD) of bulk-1. X-ray crystal structure analysis of **bulk-1** was performed using a Bruker SMART APEX II CCD detector with Mo $K\alpha$ radiation (rotating anode, $\lambda = 0.71073 \text{ \AA}$) at 298 K. The structure was determined using direct methods (SIR2004)⁵¹ and refined by full-matrix least-squares refinement on F^2 (SHELXL-2013)⁵² using the CrystalStructure software package⁵³. Crystallographic data for **bulk-1** ($\text{C}_{14}\text{H}_{10}\text{FeN}_6\text{Pt}$): Mr = 513.21, crystal dimensions $0.20 \times 0.20 \times 0.05 \text{ mm}^3$, orthorhombic, space group Cmmm (no. 65), $a = 7.5075(8) \text{ \AA}$, $b = 15.2605(16) \text{ \AA}$, $c = 7.4436(8) \text{ \AA}$, $V = 852.80(16) \text{ \AA}^3$, $Z = 2$, $d_{\text{calc}} = 1.999 \text{ g/cm}^3$, $F(000) = 480.00$, $\mu = 9.047 \text{ mm}^{-1}$, $T = 298(2) \text{ K}$. A total of 4,737 reflections were collected, of which 587 reflections were unique ($R_{\text{int}} = 0.0176$). The refinement of 41

parameters led to the final $R_1 = 0.0072$ ($I > 2\sigma(I)$), $wR_2 = 0.0173$ (all data), goodness of fit (GOF) = 1.014, and residual electron density max/min = 0.243/−0.455 eÅ^{−3}. CCDC 1405911 contains further information and details of crystallographic data for **bulk-1**. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif.

Synchrotron X-ray powder diffraction (XRPD) measurements. XRPD measurements of **bulk-1** were performed using a large imaging plate Debye–Scherrer camera installed on the BL02B2 beamline of the SPring-8, Japan. The incident X-ray was monochromated to 8 keV ($\lambda = 1.550$ Å) with a Si(111) double crystal monochromator. The samples were sealed in the glass capillaries, and all the XRPD patterns were obtained with a 0.01° step at rt.

Sorption experiments on bulk-1. The sorption experiments for water (H₂O), methanol (MeOH), ethanol (EtOH), acetonitrile (MeCN), 2-propanol, chloroform, benzene (298 K), N₂ (77 K) and CO₂ (195 K) were performed using a BELSORP-max apparatus (MicrotracBEL Corp.). The as-synthesised **bulk-1** (approximately 110 mg) was dried under high vacuum (< 10^{−2} Pa) at 343 K for 12 hours before each measurement.

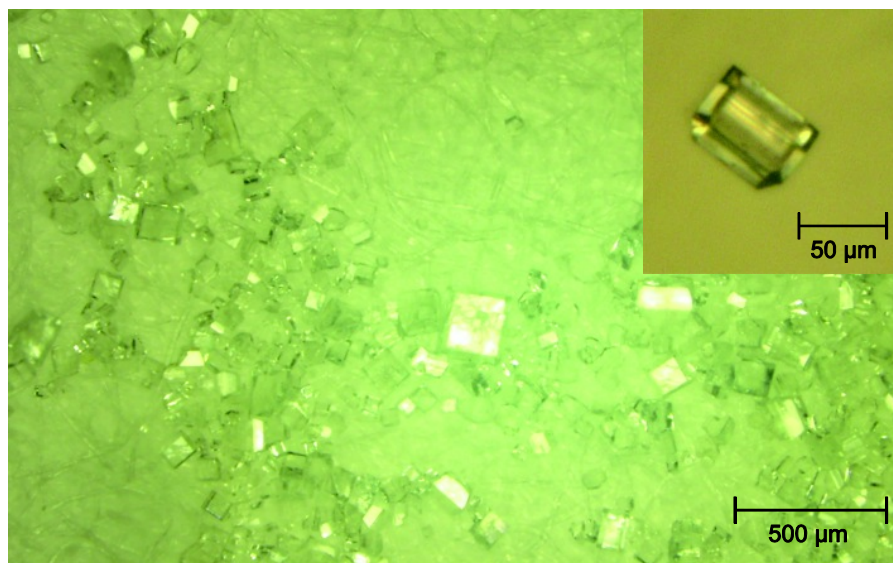
Preparation of film-1-30L. The starting material, [(C₄H₉)₄N]₂[Pt(CN)₄], was obtained from the cation-exchange reaction of K₂[Pt(CN)₄] using [(C₄H₉)₄N](ClO₄) in methanol. Fe(BF₄)₂·6H₂O, and py were used as purchased. First, the Au/Cr/Si substrate was soaked in an ethanol (EtOH) solution of 4-mercaptopyridine overnight. After this treatment, the substrate was washed with EtOH and dried by N₂ flow. Next, the substrate was alternately soaked in two types of EtOH solutions of Fe(BF₄)₂·6H₂O (10 mM) and [(C₄H₉)₄N]₂[Pt(CN)₄] (10 mM) containing excess amount of py (100 mM), in this order for 3 min. at rt for a total of 30 cycles. In this work, we also varied the cycle numbers up to 150. By repeating this simple LbL protocol, thin films of **1** with various thicknesses (**film-1-60L**, **film-1-120L**, and **film-1-150L**) were successfully obtained. The substrate was washed with EtOH for 1 min. after each soaking.

Surface XRD measurements. The preliminary surface XRD measurements on obtained thin film samples were performed using a Rigaku SmartLab X-ray diffractometer with Cu K α radiation (rotating anode, $\lambda = 1.5418$ Å). For the variable temperature measurements (Supplementary Fig. S9), a dome-type high temperature attachment (DHS-900, Anton Parr) was also used to control the temperature.

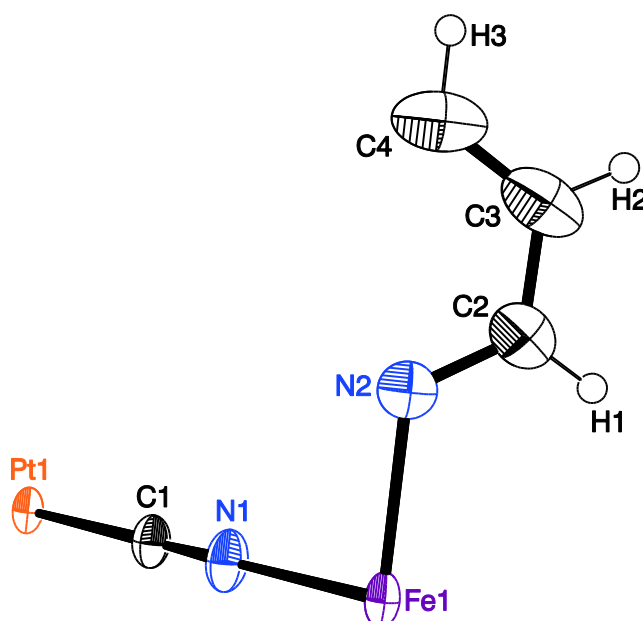
Synchrotron surface XRD measurements. Synchrotron XRD measurements on the obtained thin film samples were performed using a multi-axis diffractometer (Kohzu-Seiki TDT-17) with a NaI scintillation detector installed on the BL13XU beamline of SPring-8, Japan. The incident X-ray was monochromated to 8 keV ($\lambda = 1.550 \text{ \AA}$) with a Si111 double-crystal monochromator. In-plane and out-of-plane XRD patterns were obtained with a 0.1° step. Out-of-plane XRD measurement was performed in a typical θ - 2θ scattering geometry. Rocking-curve (θ -scan mode) and azimuthal-angle-dependence (ϕ -scan mode) measurements were performed at the 020 peak position in the out-of-plane geometry (Supplementary Fig. S13) with 0.1° and 1° steps, respectively. For the in-plane XRD (grazing-incidence (GI) XRD mode), strong diffraction can be observed on the condition that the X-ray incident angle (α) to the sample is below a critical angle. Thus, an incident-angle dependence measurement at the 001 peak position was performed before the in-plane XRD measurement, and the diffraction pattern was collected at $\alpha = 0.15^\circ$. A Soller slit (Huber 3030-I, 0.4°) was placed between the sample and detector to improve the signal-to-noise (S/N) ratio of the diffraction. Both in-plane and out-of-plane XRD measurements were performed under constant He flow. The XRD profile fittings shown in Fig. 3 were performed using the Topas program⁵⁴. The simulated XRD patterns shown in Fig. 3 were obtained using the Mercury software package⁵⁵ based on the perfectly oriented crystal model of **bulk-1**. For the *in situ* XRD measurements under vapours of several solvents, the vapour pressure in He gas was controlled by MFC, bubbler, and condenser installed on a BEL-Flow apparatus (MicrotracBEL Corp.).

Molecular mechanics calculations. Ideally guest-adsorbed model structures were obtained from molecular mechanics calculations installed on HyperChem Pro ver. 8.0.10 (HyperCube Inc.). The basic framework structure of **1** was generated from its single crystal X-ray structure and used as its expanded structure ($3 \times 1 \times 3$ supercell) without symmetry restriction. The lattice constants were set to the results of the *in situ* synchrotron surface XRD measurements under the vapour pressure of $P/P_0 = 0.93$. For geometry optimisation (MM+)⁵⁶ of the guest molecules, the framework structures were fixed, and the PolakRibiere optimiser was used (RMS Gradient: $0.001 \text{ kcal/ \AA mol}$).

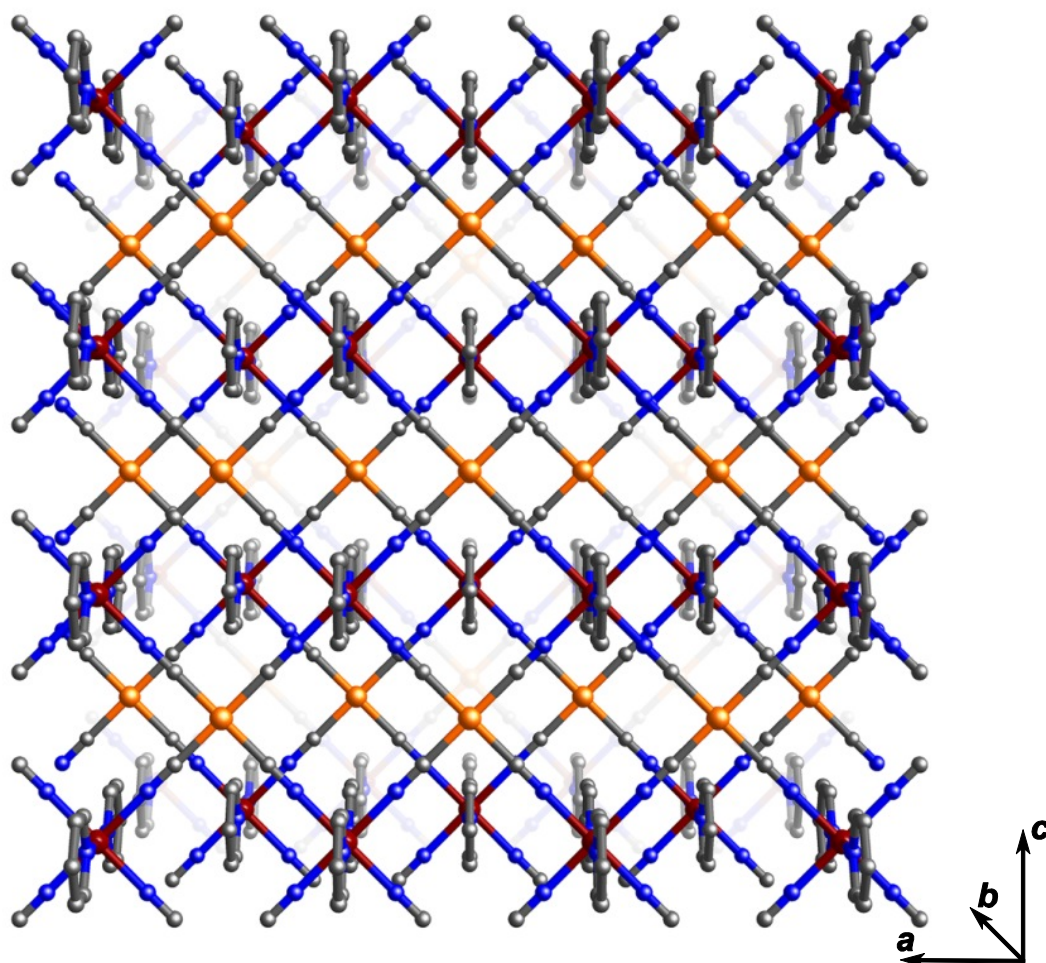
2. Supplementary Figures and Table



Supplementary Figure 1 | A photograph of single crystals of bulk-1. (Inset) A typical example of the block-shaped single crystal for single-crystal X-ray diffraction.



Supplementary Figure 2 | ORTEP drawing of bulk-1 (50% probability).

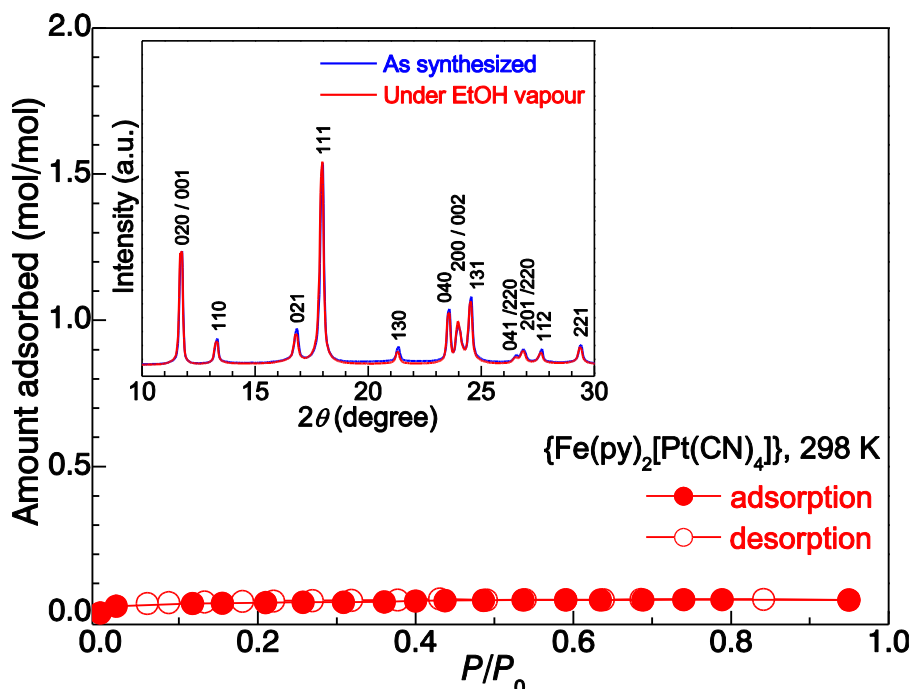


Supplementary Figure 3 | 2D layer stacking structure of bulk-1 viewed along *b*-axis. C = grey, N = blue, Fe = red, Pt = orange. H atoms are omitted for clarity.

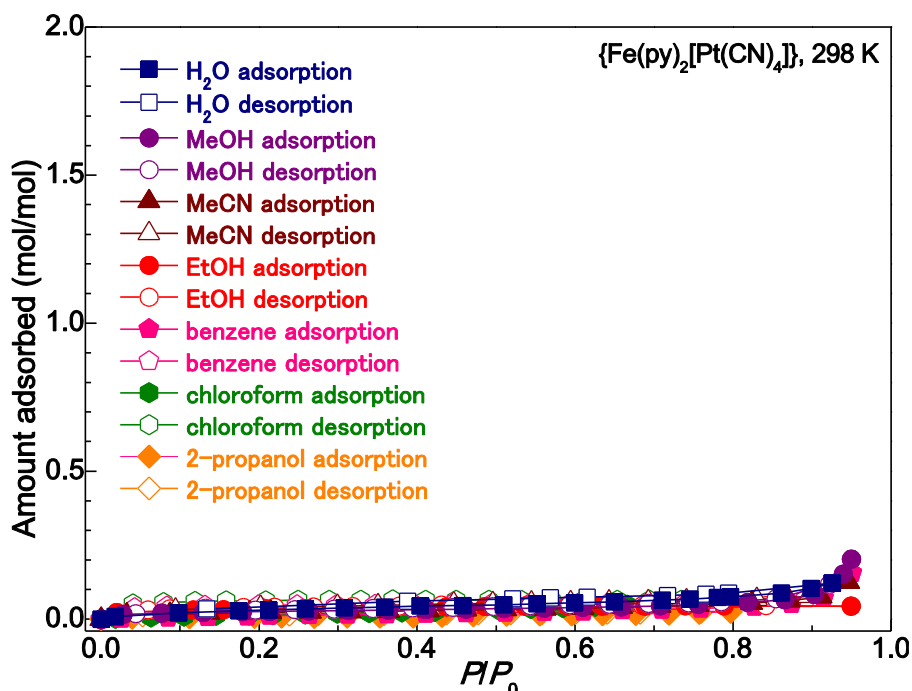
Supplementary Table S1 | Crystal data and structure refinement of bulk-1.

Empirical Formula	C ₁₄ H ₁₀ Fe N ₆ Pt
Formula weight	513.21
Temperature	298(2) K
Wavelength	0.71073 Å
Crystal system	orthorhombic
Space group	Cmmm
Unit cell dimensions	$a = 7.5075(8) \text{ Å}, \alpha = 90^\circ$ $b = 15.2605(16) \text{ Å}, \beta = 90^\circ$ $c = 7.4436(8) \text{ Å}, \gamma = 90^\circ$
Volume (V)	852.80(16) Å ³
Z	2
Density (calculated)	1.999 g/cm ³
Absorption coefficient	9.047 mm ⁻¹
<i>F</i> (000)	480
Crystal size	0.20 × 0.20 × 0.05 mm ³
Theta range for data collection	2.669 to 27.534°
Index ranges	$-9 \leq h \leq 9, -19 \leq k \leq 19, -9 \leq l \leq 9$
Reflections collected	4,743
Independent reflections	587 ($R_{\text{int}} = 0.0176$)
Completeness to theta = 27.534°	99.8%
Absorption correction	Multi-scan
Max. and min. transmission	0.660 and 0.573
Refinement method	Full-matrix least-squares on F^2
Data/restraints/parameters	587/0/41
Goodness-of-fit on F^2	1.014
Final <i>R</i> indices [$I > 2\sigma(I)$]	$R_1 = 0.0072, wR_2 = 0.0173$
<i>R</i> indices (all reflections)	$R_1 = 0.0072, wR_2 = 0.0173$
Largest diff. peak and hole	0.243 and -0.455 e Å^{-3}
CCDC reference number	1405911

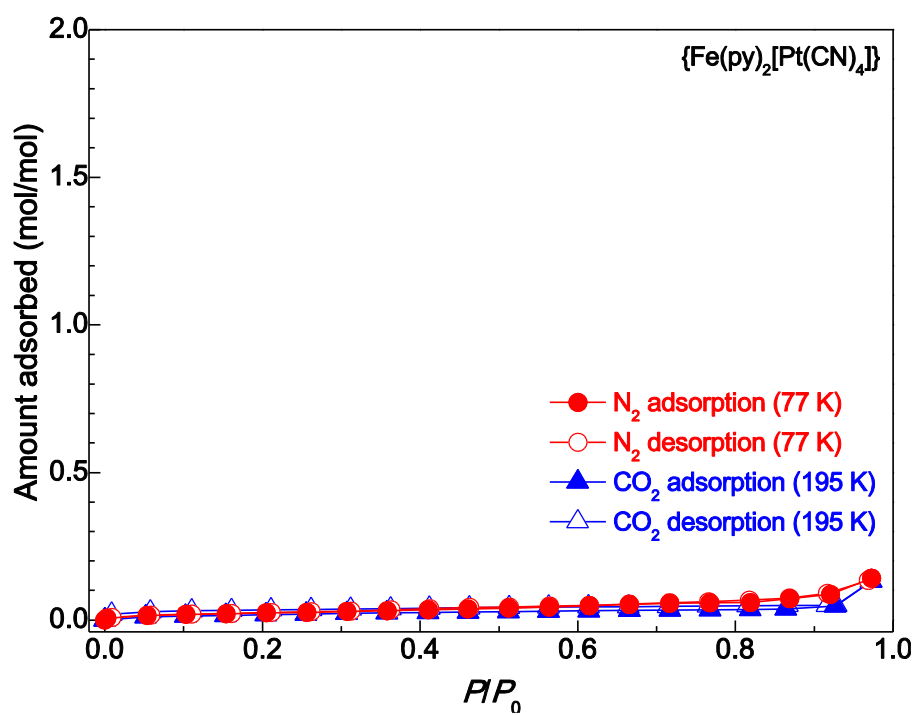
***CheckCIF/PLATON information:** From checkCIF/PLATON report, any levels A, B, and C alerts were not suggested.



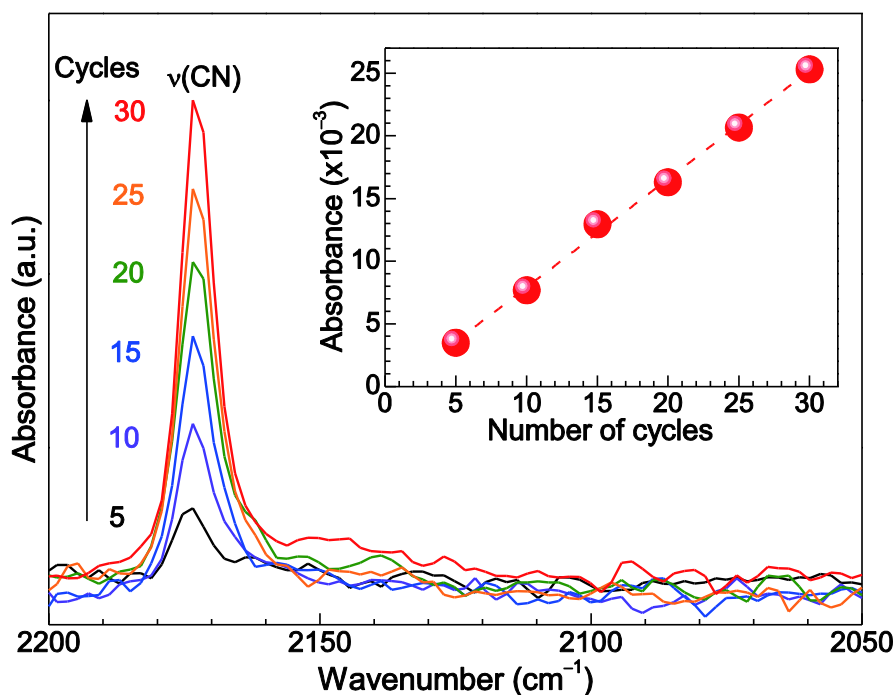
Supplementary Figure 4 | EtOH sorption isotherm of bulk-1 at 298 K. As clearly observed, **bulk-1** shows no EtOH uptake. Inset shows synchrotron XRPD profiles ($\lambda = 1.550 \text{ \AA}$) of as synthesized **bulk-1** (blue) and **bulk-1** under saturated vapour of EtOH (red).



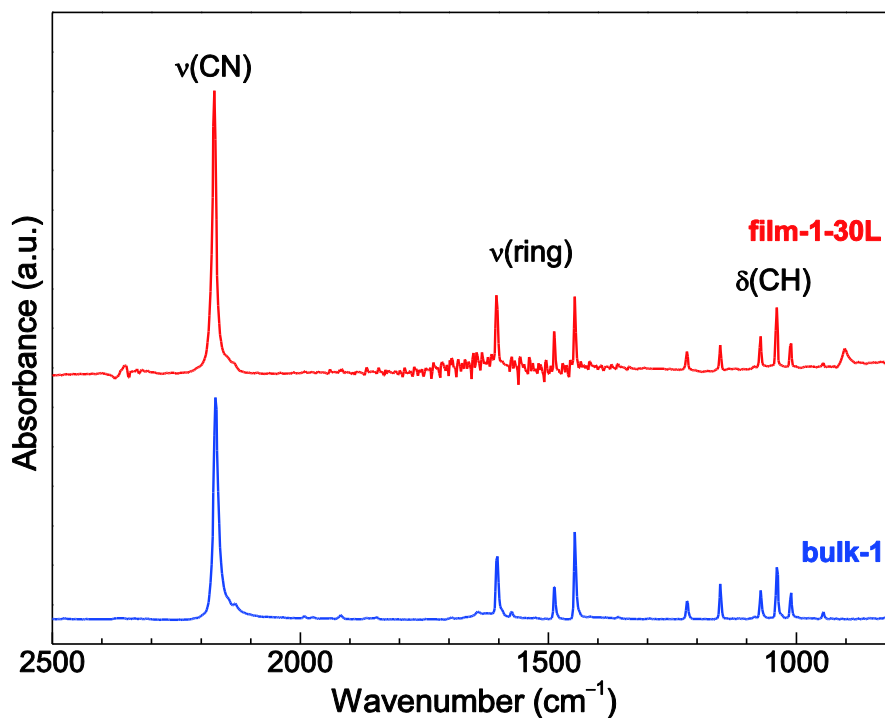
Supplementary Figure 5 | Adsorption isotherms of bulk-1 for several solvent vapours at 298 K. As clearly seen, **bulk-1** shows no significant guest uptake.



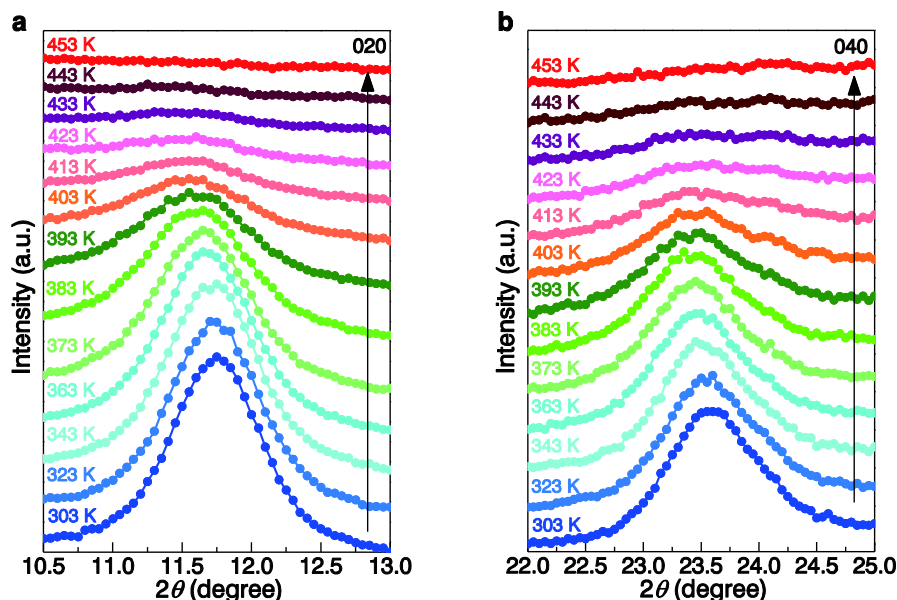
Supplementary Figure 6 | Adsorption isotherms of bulk-1 for N_2 (77 K, circles) and CO_2 (195 K, triangles).



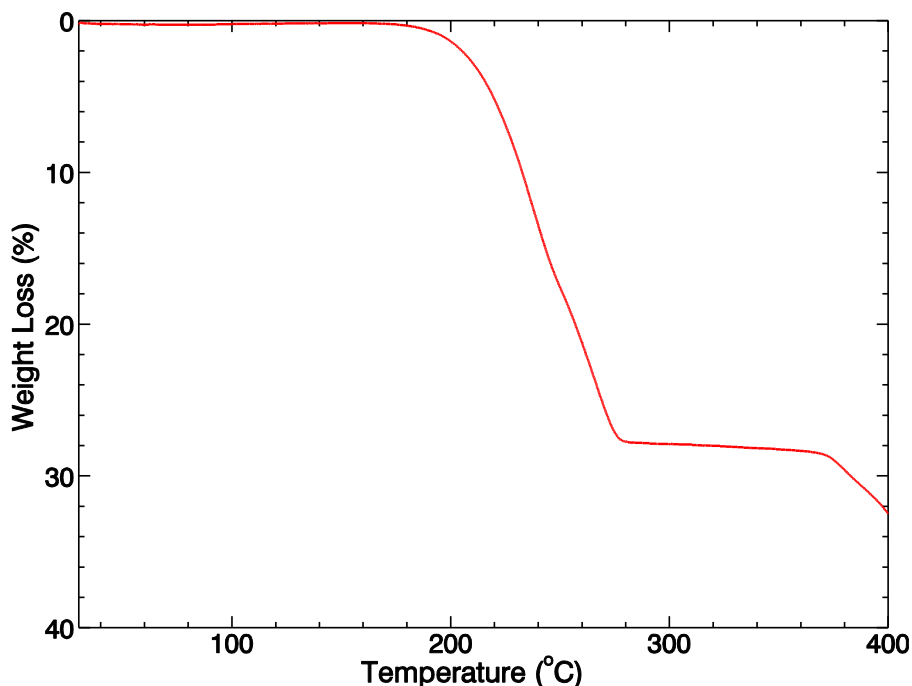
Supplementary Figure 7 | LbL thin-film growth monitored by IRRAS at rt. (Inset) Dependence of the maximum absorbance of the $\nu(\text{CN})$ mode on the number of cycles. The red dotted line is a visual guide.



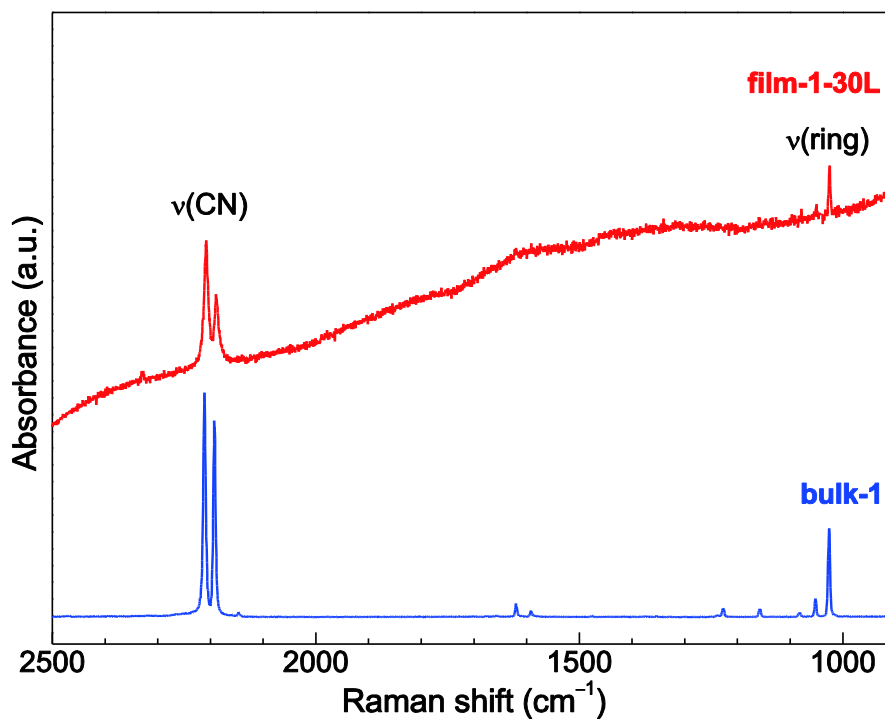
Supplementary Figure 8 | IR spectra of bulk-1 (blue) and film-1-30L (red) at rt.



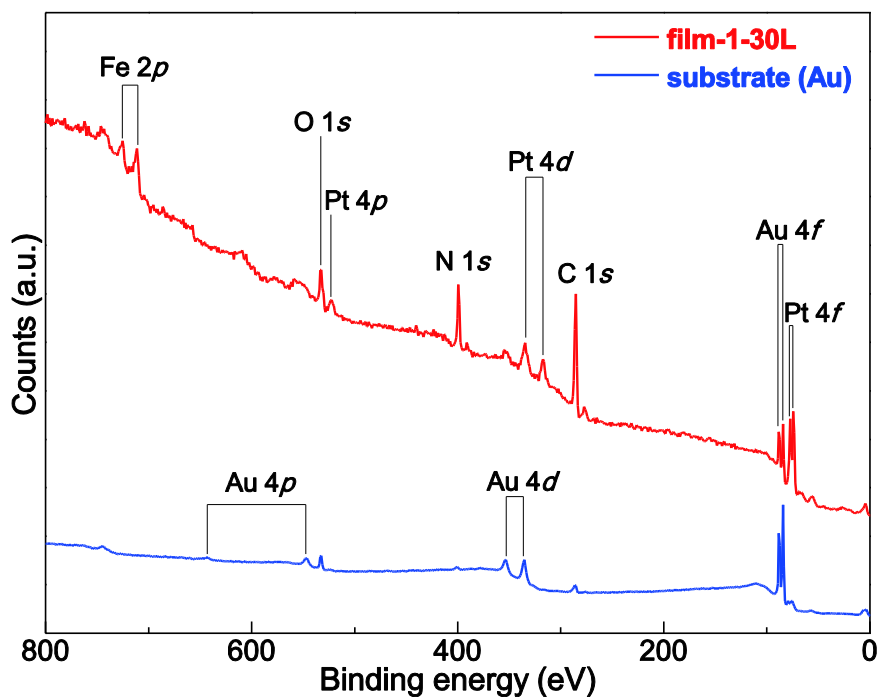
Supplementary Figure 9 | Thermal stability of film-1-30L. a, b, Out-of-plane XRD profiles of **film-1-30L** in 020 (a) and 040 (b) region. Decomposition of the thin film occurred at approximately 423 K. The decomposition temperature of the thin film is lower than that of **bulk-1** (see Supplementary Fig. S10).



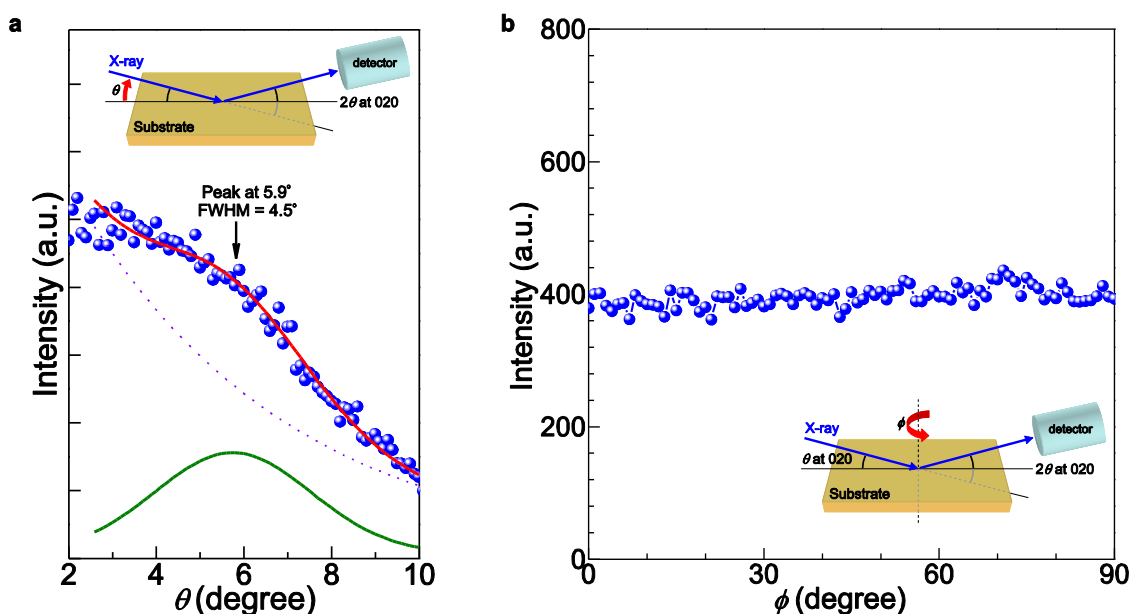
Supplementary Figure 10 | Thermal stability of bulk-1. Based on the TG measurement, decomposition of **bulk-1** occurred at approximately 470 K.



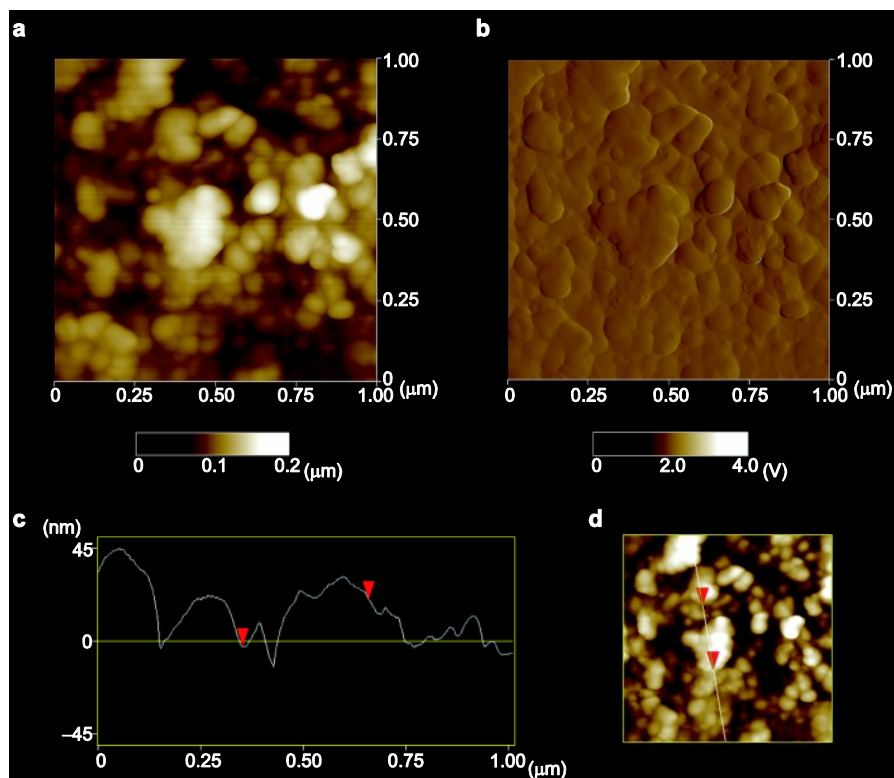
Supplementary Figure 11 | Raman spectra of bulk-1 (blue) and film-1-30L (red) at rt.



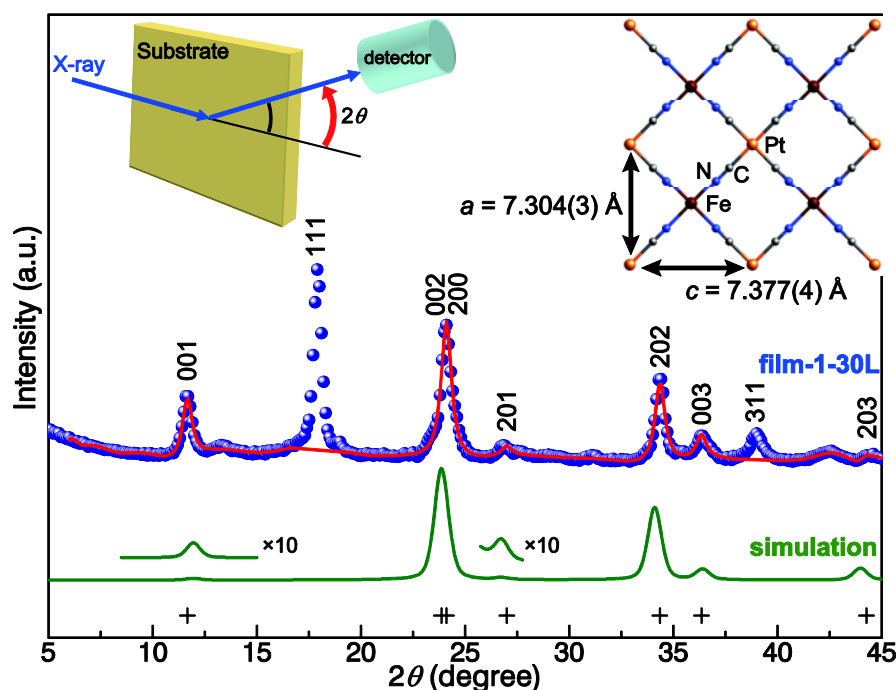
Supplementary Figure 12 | XPS spectra of Au substrate (blue) and film-1-30L (red) at rt. The peak assignments are also shown.

**Supplementary Figure 13 | Confirmations of tilted nature and uniformity of film-1-30L.**

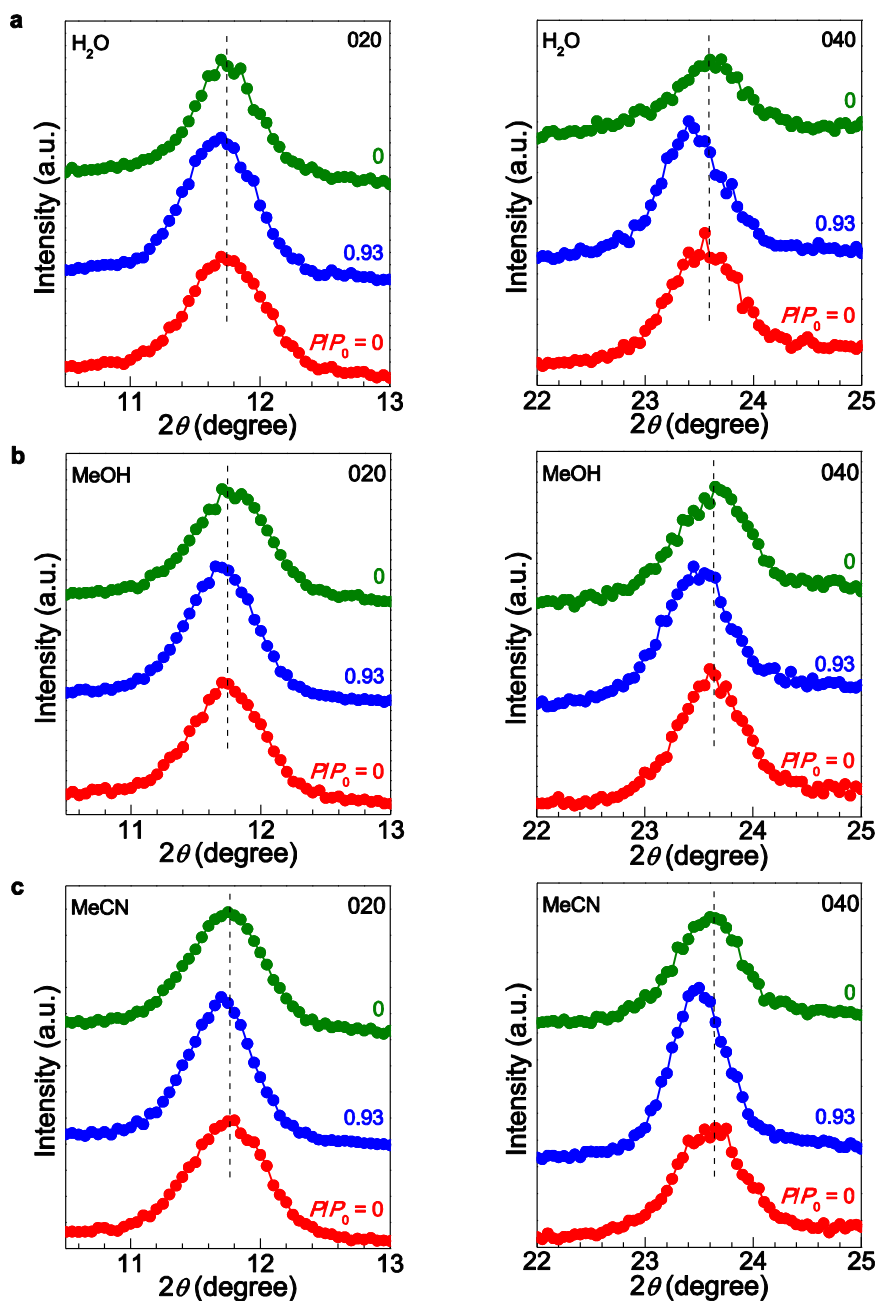
Rocking-curve (a) and azimuthal-angle-dependence (b) measurements at 020 position. The blue circles in each panel and the red and green lines in panel a denote the experimental data, fitting result, and background subtracted fitting diffraction patterns, respectively. The inset figures in each panel show the experimental geometries.



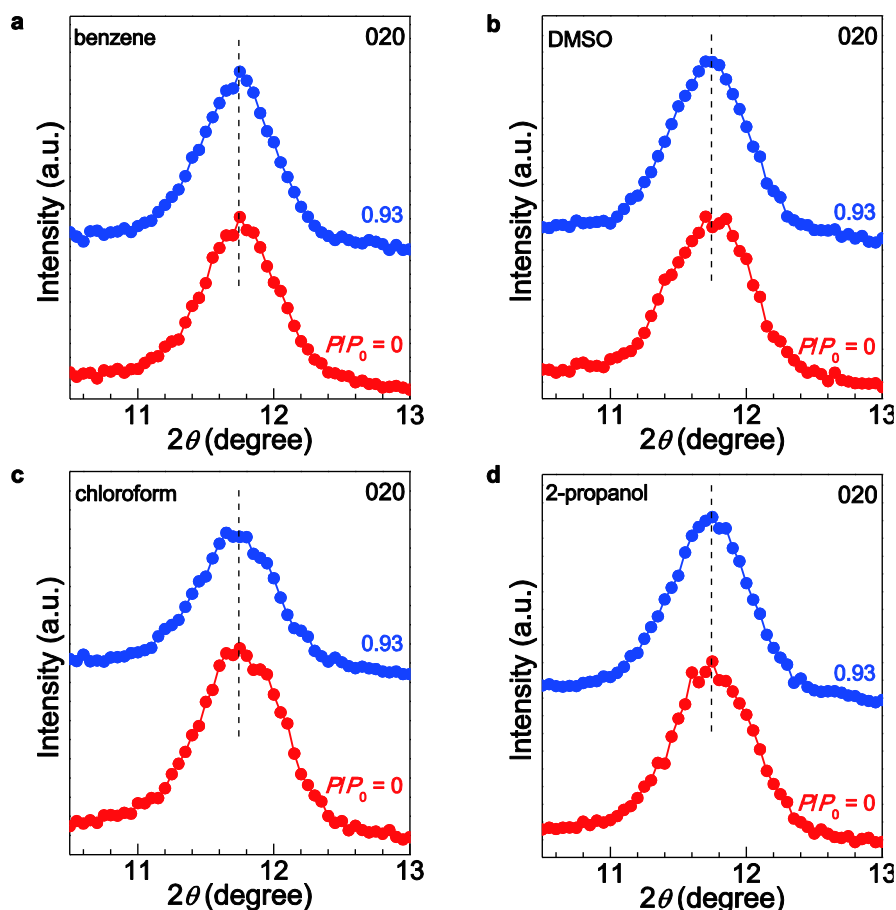
Supplementary Figure 14 | AFM observation of film-1-30L at rt. **a, b**, 2-D images of height mode (**a**) and amplitude mode (**b**). **c**, Line profile of height image. The profile was recorded along the white line shown in panel **d**. The average thickness calculated from the out-of-plane XRD using Scherrer's equation (approximately 16 nm) is consistent with this observation. On the other hand, the crystal domain size of this film estimated from the in-plane XRD using Scherrer's equation was approximately 22 nm. From AFM image, several domains (islands) could be observed of this thin film. Some large domains had a side of 70 – 100 nm, while other small domains had a side of less than 30 nm. Therefore, because of the large size dispersion of the domains, the crystal domain size would be estimated as a small value in the X-ray observation.



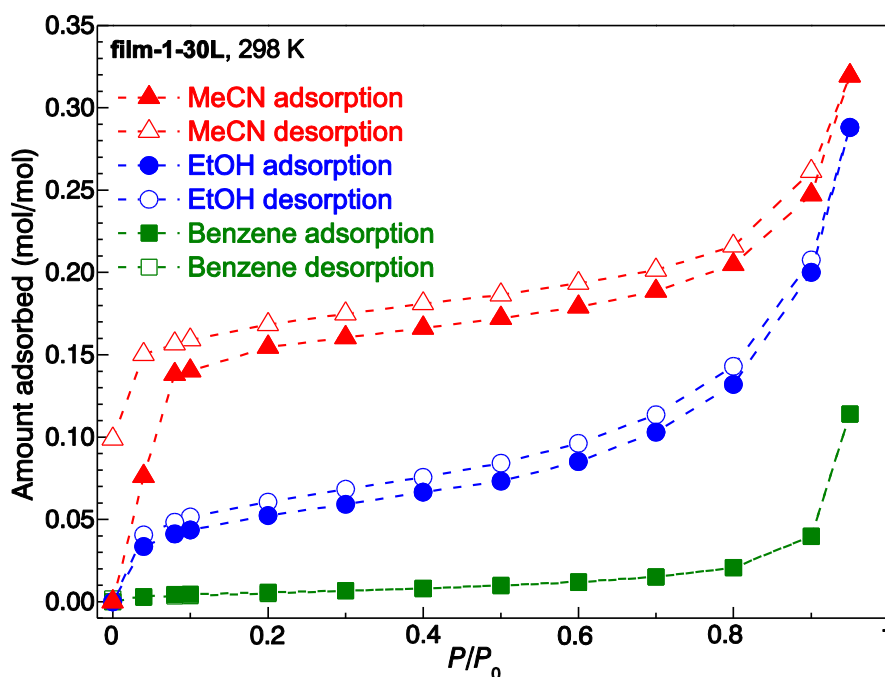
Supplementary Figure 15 | In-plane XRD profile fitting of film-1-30L under EtOH vapour ($P/P_0 = 0.93$, $\lambda = 1.550$ Å). The blue circles, red lines, green lines and black crosses denote the experimental data, fitting results, simulated patterns, and Bragg peaks, respectively. (orthorhombic, Cmmm, $a = 7.304(3)$, $c = 7.377(4)$ Å). As clearly seen, no significant change was observed in in-plane diffraction pattern showing anisotropic structural change of **film-1-30L** under EtOH vapour.



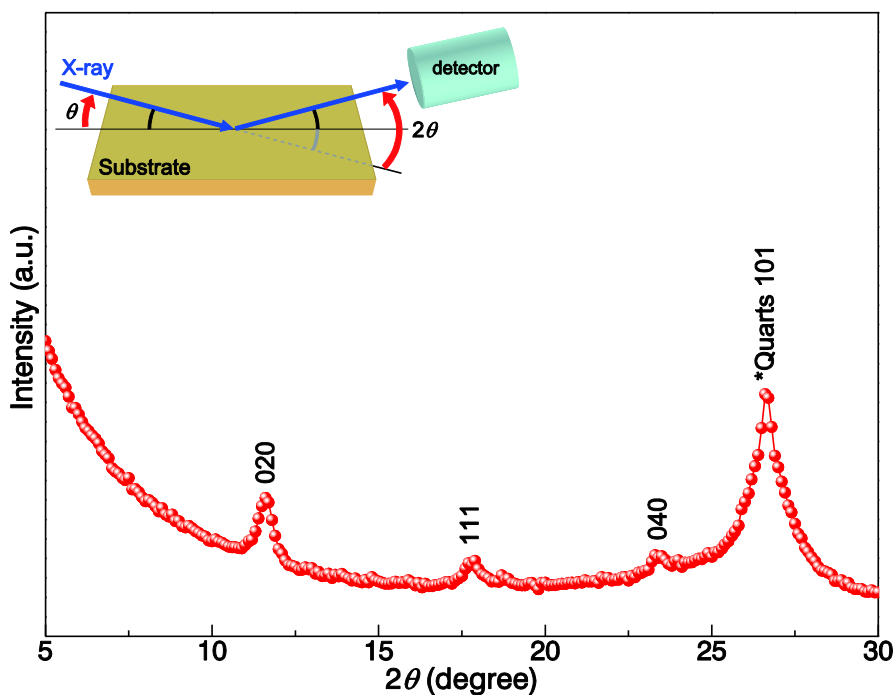
Supplementary Figure 16 | Observations of XRD profile ($\lambda = 1.550 \text{ \AA}$) under different molecular species (part I). a, b, c, *In situ* out-of-plane XRD profiles (020 (left panels) and 040 (right panels) regions) of **film-1-30L with varying PIP_0 of water (a), MeOH (b), and MeCN (c) vapours. For relatively small guest molecules, the reversible peak shifts were also observed.**



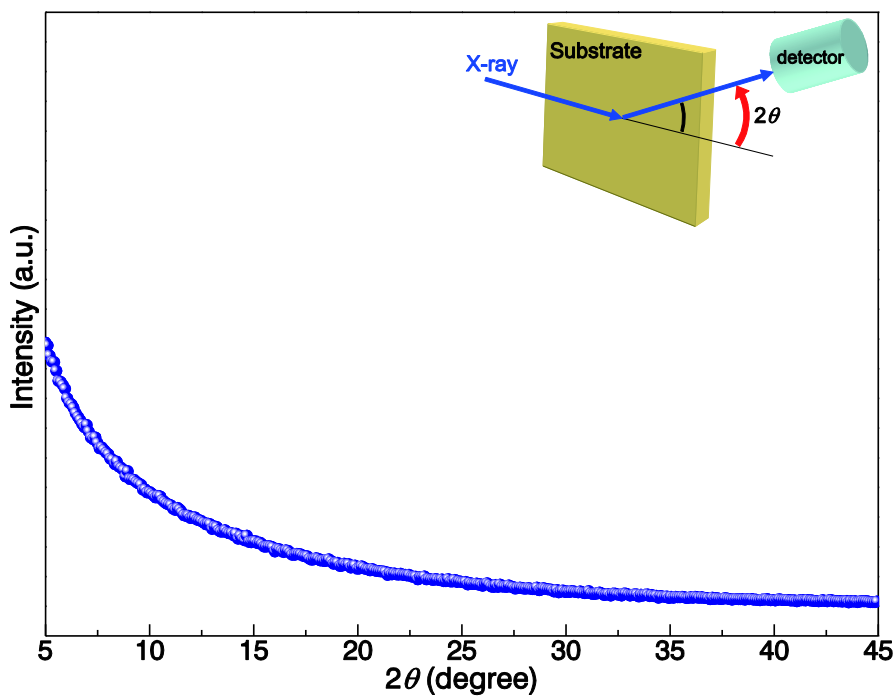
Supplementary Figure 17 | Observations of XRD profile ($\lambda = 1.550 \text{ \AA}$) under different molecular species (part II). a, b, c, d, *In situ* out-of-plane XRD profiles (020 region) of **film-1-30L** with varying P/P_0 of benzene (a), dimethyl sulfoxide (DMSO) (b), chloroform (c), and 2-propanol (d) vapours. For relatively large guest molecules, no significant change in the XRD profile was observed.



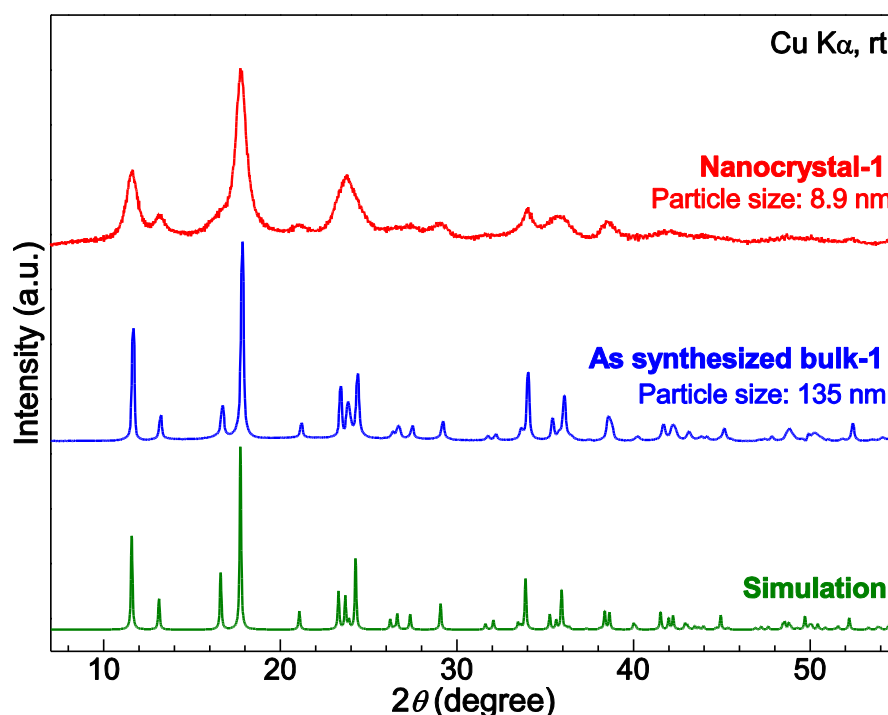
Supplementary Figure 18 | Sorption isotherms for EtOH (circles), MeCN (triangles) and benzene (squares) of film-1-30L obtained from QCM measurements at 298 K. For EtOH and MeCN, clear uptakes were observed. Although clear structural transformations with hysteresis are observed in *in situ* XRD measurements, their adsorption behaviours with slight hysteresis are not so clear compared with ideal gate-opening-type adsorption behaviour. This would be due to the difference in the substrates. For QCM measurements, thin film samples fabricated on commercially available Au electrode on a quartz tip without any annealing procedures were used. Because of the setup limitation for QCM equipment, Au on QCM tip has relatively rough surface compared with the Au/Cr/Si substrate for XRD measurements. Supplementary Figs. S19 and S20 show out-of-plane and in-plane XRD diffraction profiles of **film-1-30L** fabricated on QCM tip. For out-of-plane geometry, clear 020 and 040 diffraction peaks of **1** were observed, whereas any diffraction peaks were not detected in in-plane geometry. This means that average crystal domain size along in-plane (parallel to the QCM tip) direction is too small to be detected, or thin film does not have clear periodic order along in-plane direction, which are derived from the rough surface nature of Au on QCM tip. Therefore, it is suggested that the sorption isotherms obtained from QCM observe the summation of the guest uptakes of the thin film with a variety of domain sizes, resulting in the ambiguous sorption behaviour with slight hysteresis.



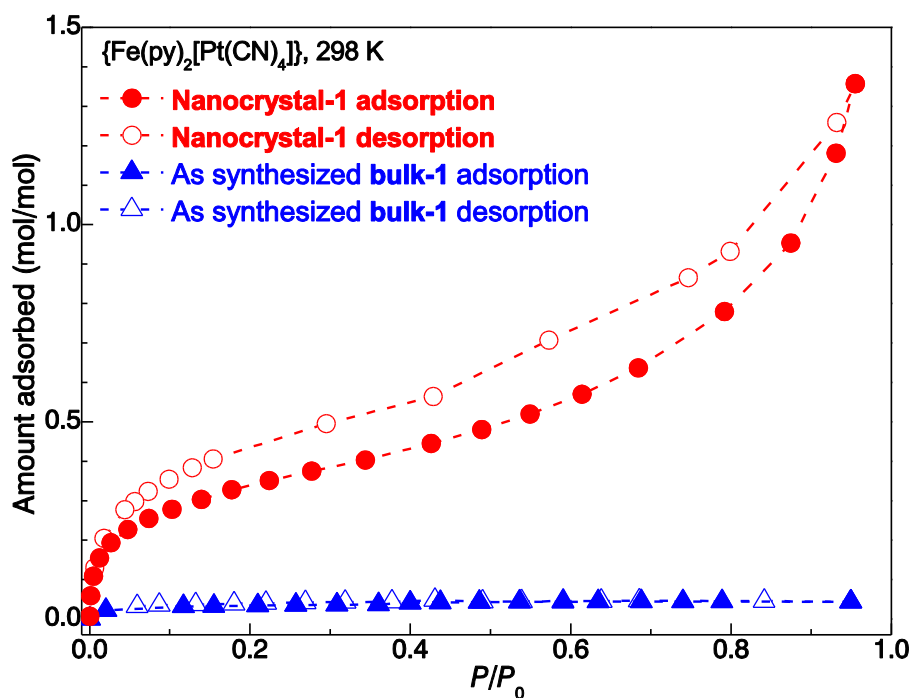
Supplementary Figure 19 | Out-of-plane XRD profile (Cu $K\alpha$, rt) of film-1-30L fabricated on a QCM tip for sorption experiments.



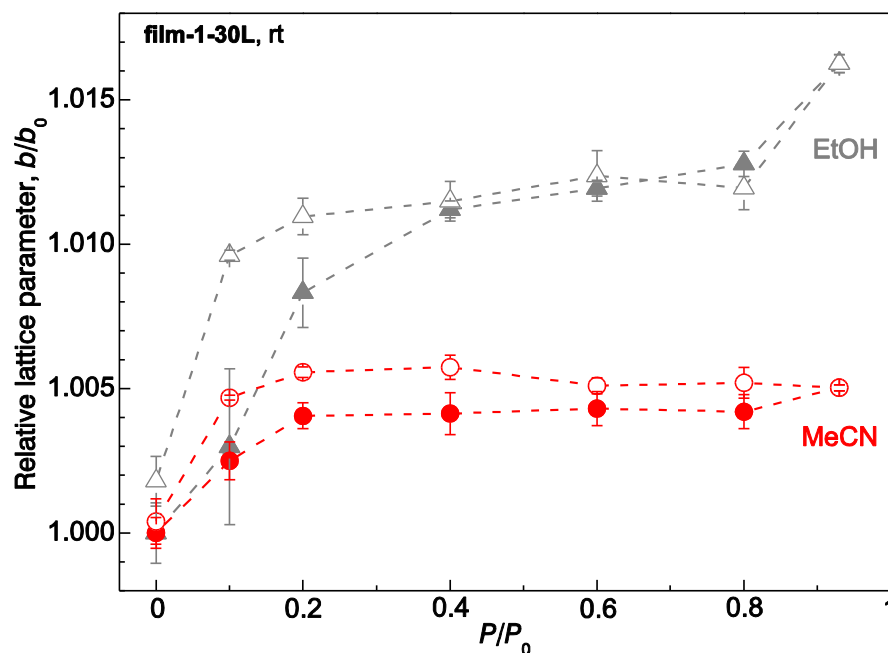
Supplementary Figure 20 | In-plane XRD profile (Cu $K\alpha$, rt) of film-1-30L fabricated on a QCM tip for sorption experiments.



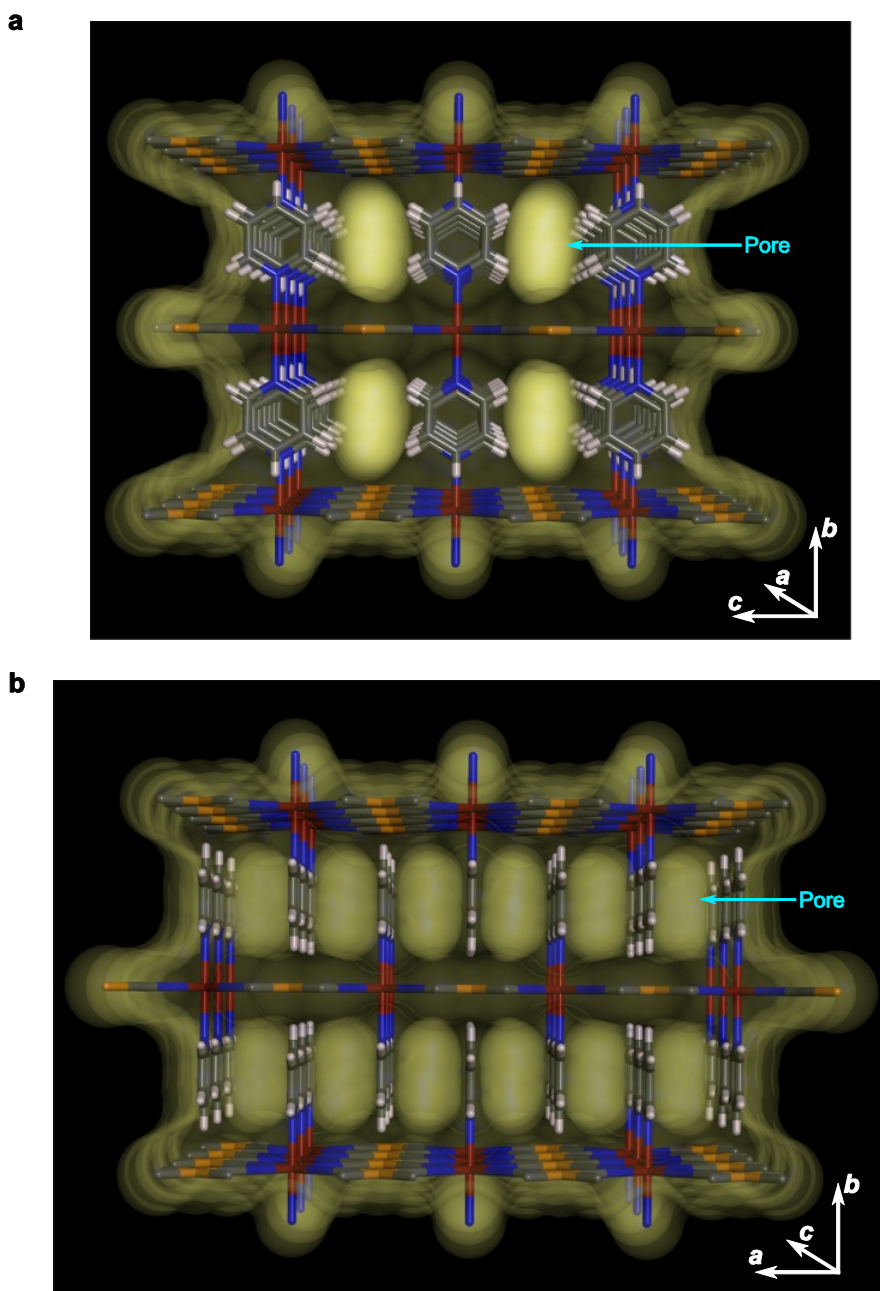
Supplementary Figure 21 | XRPD patterns of nanocrystal-1 (red), as synthesized bulk-1 (blue) and simulation of bulk-1 (green). XRPD patterns were collected using Bruker D8 ADVANCE X-ray diffractometer with Cu K α radiation at rt. For comparison, crystal downsizing effect of bulk sample was also investigated. Nanocrystal sample was prepared by mechanical grinding of as-synthesized **bulk-1** using mortar for 30 min. As clearly seen, clear broadening of diffraction pattern was observed. Crystal sizes estimated from Scherrer's equation were 8.9 nm (**nanocrystal-1**) and 135 nm (as synthesized **bulk-1**), respectively.



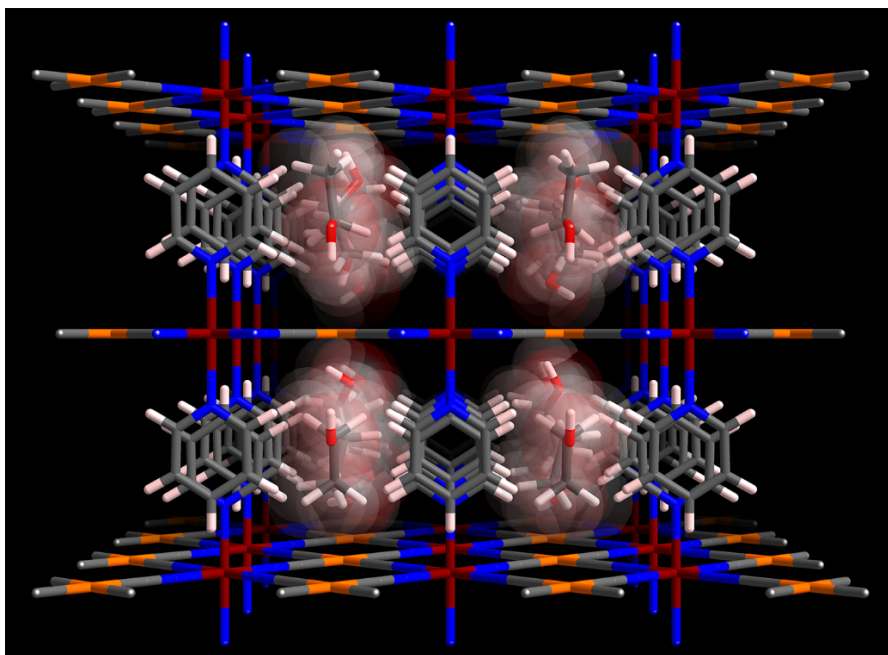
Supplementary Figure 22 | Comparison of EtOH adsorption isotherms between nanocrystal-1 (red) and as synthesized bulk-1 (blue) at 298 K. Although bulk-1 showed no EtOH uptake, nanocrystal-1 obtained from mechanical grinding showed clear EtOH sorption behaviour.

**Supplementary Figure 23 | Structural transformation of film-1-30L under MeCN vapour.**

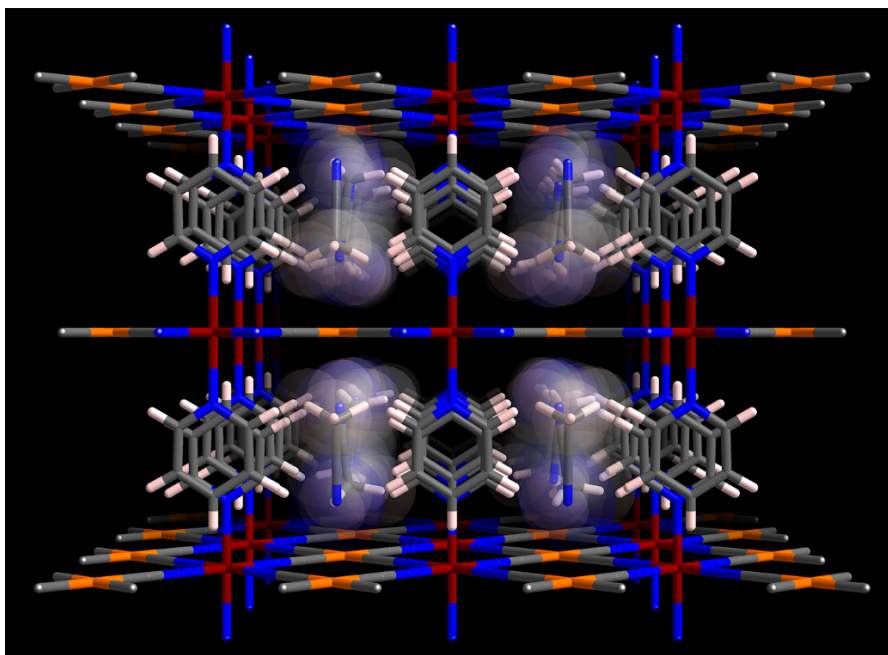
Relative lattice constant (b/b_0) calculated from 020 and 040 diffraction peak positions in *in situ* out-of-plane XRD. For comparison, the data under EtOH vapour is also shown. The error bars denote $\pm$ s.d. As clearly observed, the variation of the interlayer spacing was relatively small when MeCN vapour was introduced. Most likely, this result is due to the difference in the molecular shape of the adsorbate because the EtOH molecule is slightly bulky compared with MeCN molecule with its straight molecular shape (see Supplementary Figs. S25 and S26).



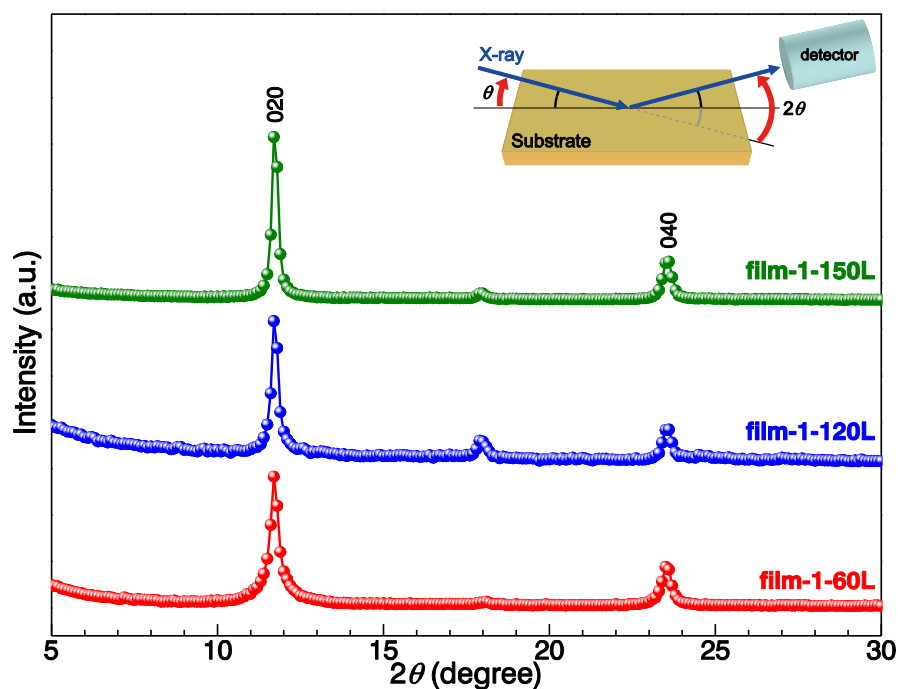
Supplementary Figure 24 | Capsule-shaped pores formed inside bulk-1 crystal. a, b, Projection of porous structure of **bulk-1** viewed along a-axis (**a**) and c-axis (**b**). C = grey, N = blue, H = pink, Fe = red, Pt = orange. In each panel, the structure is shown as stick model with superimposed solvent accessible surface (yellow part, probe radius: 1.4 Å, drawn by Discovery Studio Visualizer 4.0⁵⁷). In the crystal of **bulk-1**, a capsule-shaped pore with the dimensions of $2.9 \times 3.0 \times 5.2 \text{ Å}^3$ is formed between the 2-D layers. The total solvent accessible void volume of **1** determined by PLATON⁵⁸ is 10.4% (88.4 Å^3) per unit cell volume.



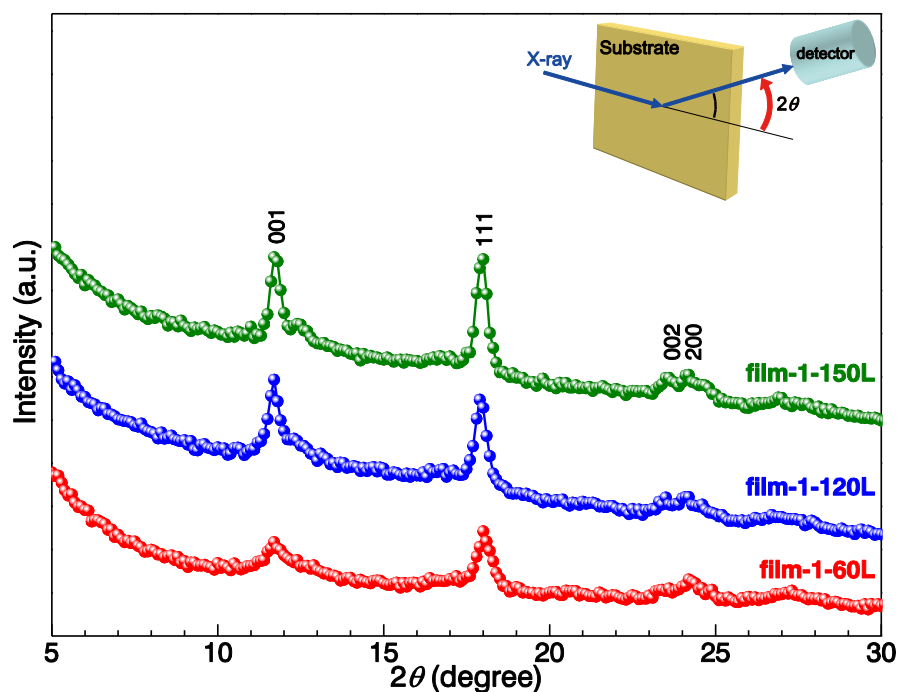
Supplementary Figure 25 | Optimised model structure (MM+) of ideally EtOH adsorbed framework. The guest EtOH molecules are shown as stick with superimposed CPK model.



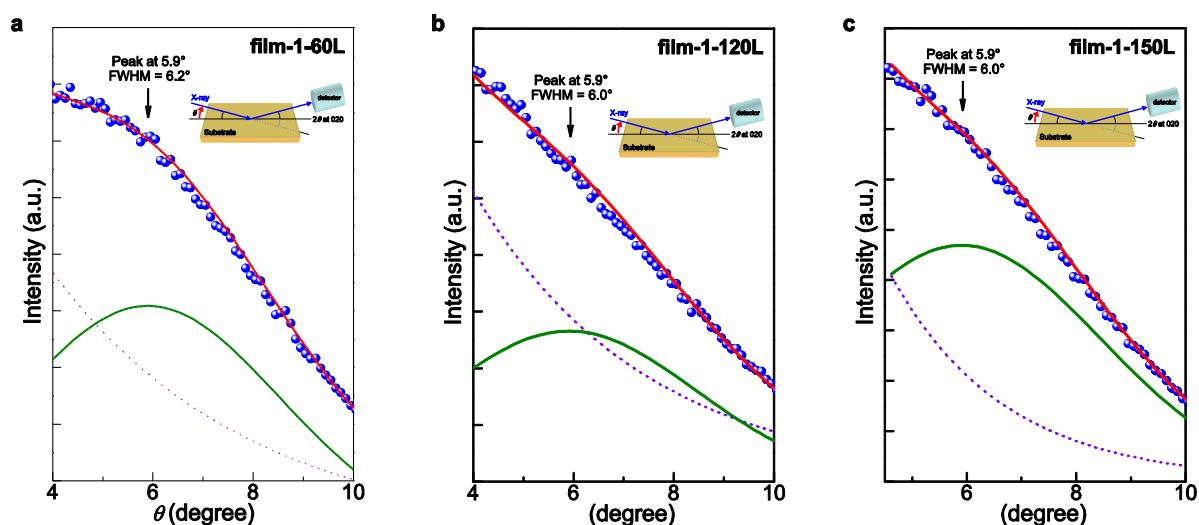
Supplementary Figure 26 | Optimised model structure (MM+) of ideally MeCN adsorbed framework. The guest MeCN molecules are shown as stick with superimposed CPK model.



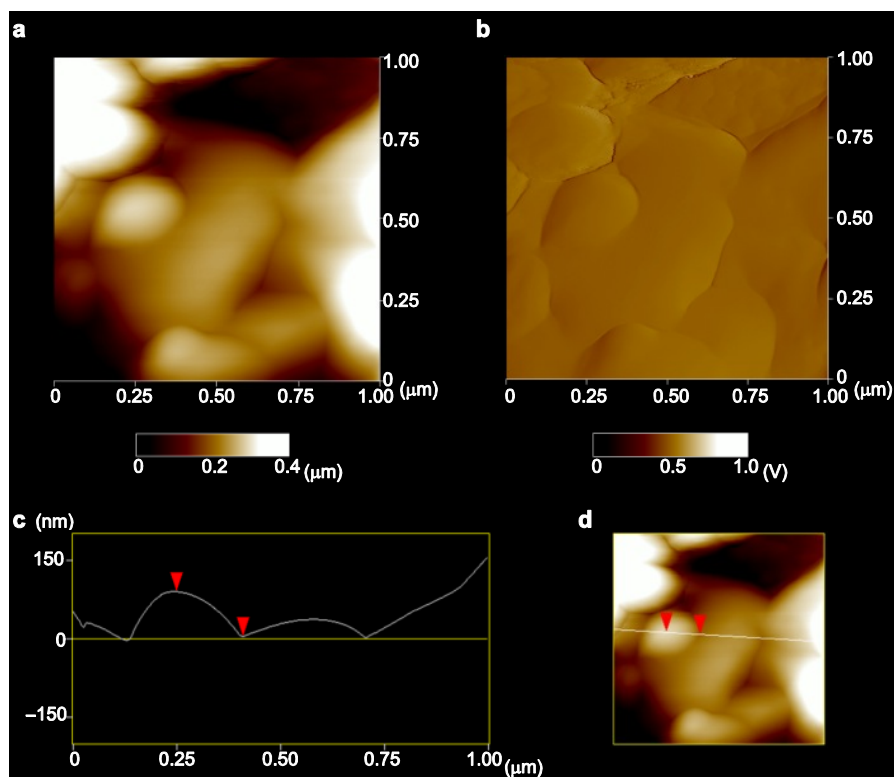
Supplementary Figure 27 | Out-of-plane XRD profiles ($\lambda = 1.550 \text{ \AA}$) of film-1-60L (red), film-1-120L (blue), and film-1-150L (green).



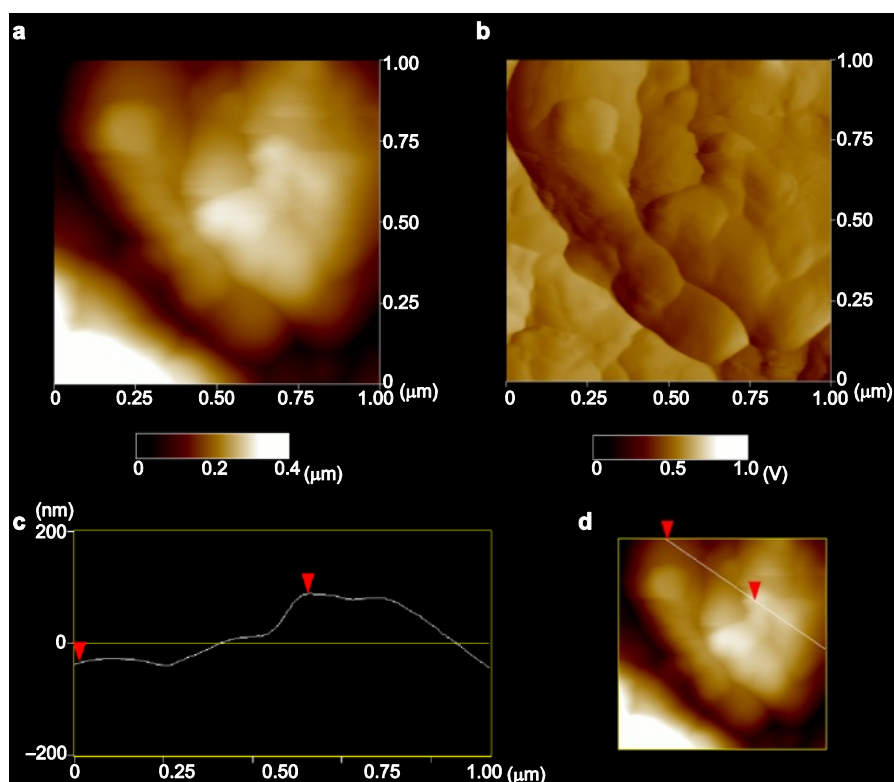
Supplementary Figure 28 | In-plane XRD profiles ($\lambda = 1.550 \text{ \AA}$) of film-1-60L (red), film-1-120L (blue), and film-1-150L (green).



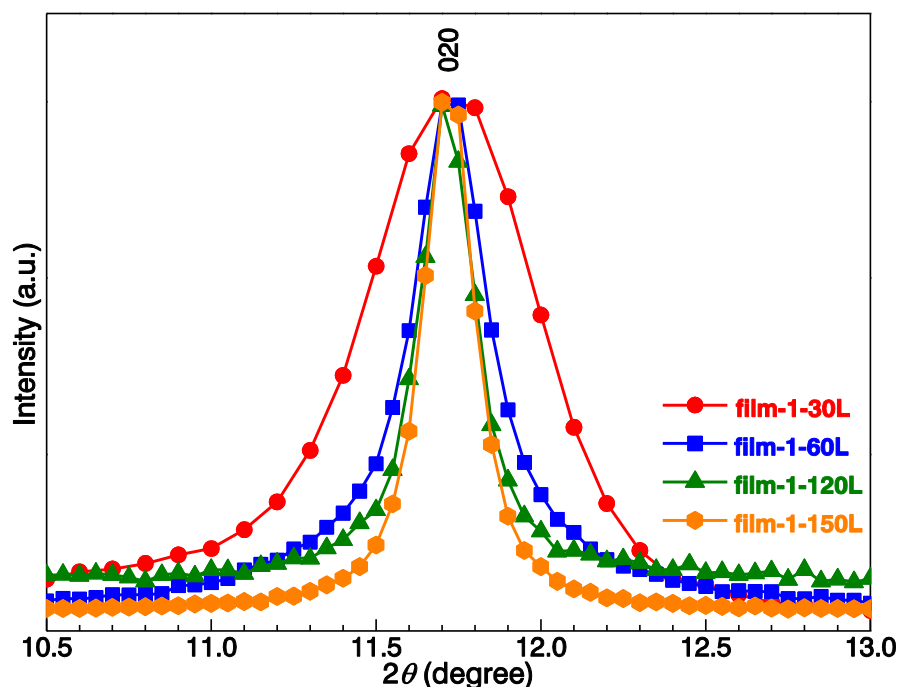
Supplementary Figure 29 | Rocking-curve measurements on film-1-60L, film-1-120L and film-1-150L. The blue circles in each panel and the red and green lines in each panel denote the experimental data, fitting result, and background subtracted fitting diffraction patterns, respectively. The inset figures in each panel show the experimental geometry. The average tilting angle estimated from fitting analysis is almost constant (approximately 6°), slightly increasing from that of **film-1-30L** (4.5°), while experimental rocking curve profile becomes vaguer with increasing the number of LbL cycle. This would be due to large size and tilting angle dispersion by addition of more layers.



Supplementary Figure 30 | AFM observation of film-1-60L (red) at rt. **a, b**, 2-D images of height mode (**a**) and amplitude mode (**b**). **c**, Line profile of height image. The profile was recorded along the white line shown in panel **d**. For film-1-60L, clear AFM image was collected, where thickness and the domain (island) size are remarkably increased compared with those of **film-1-30L** (Supplementary Fig. S14). Estimated domain sizes from AFM image are as follows: 50 – 75 nm (thickness) and > 120 nm (side).

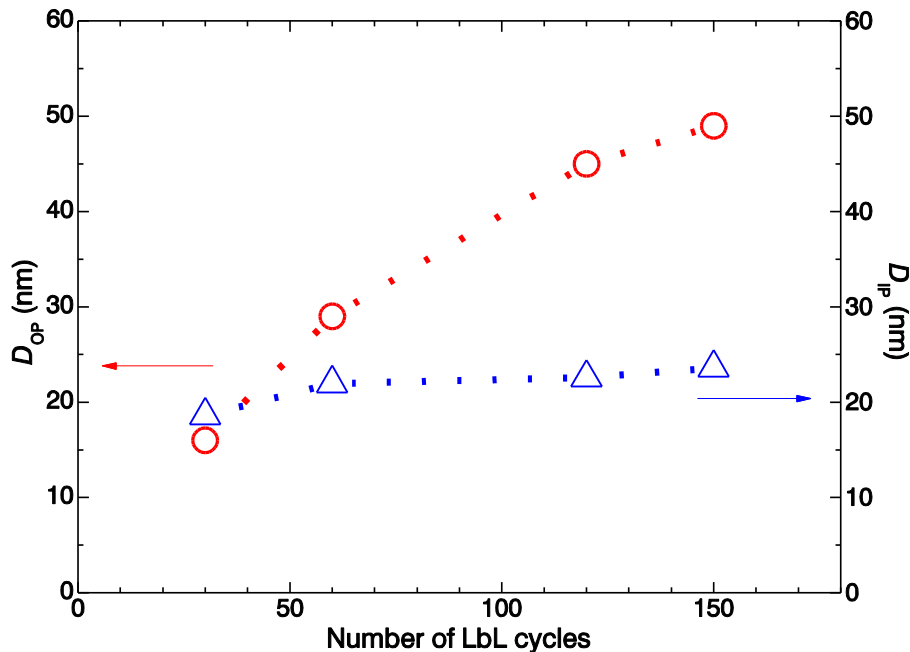


Supplementary Figure 31 | AFM observation of film-1-90L at rt. **a, b**, 2-D images of height mode (**a**) and amplitude mode (**b**). **c**, Line profile of height image. The profile was recorded along the white line shown in panel **d**. For **film-1-120L** and **film-1-150L**, AFM observations were not possible because of experimental difficulty derived from the large thickness and softness of MOF film. Therefore, for comparison, we fabricated similar thin film using 90 LbL cycles (**film-1-90L**) only for this AFM observation. As clearly observed, thickness and the domain (island) size are increased compared with those of **film-1-60L**. From the AFM observations of **film-1-60L** and **-90L**, the island size along the parallel direction relative to the substrate seems to become larger and interfaces among adjacent domains seem to become more unclear with increasing the LbL cycle. Although the calculated domain (island) size from in-plane XRD is almost constant (~ 22 nm), island morphology and surface structure are clearly changed via increase in the LbL cycle. Estimated domain sizes from AFM image are as follows: 100 – 130 nm (thickness) and > 150 nm (side).

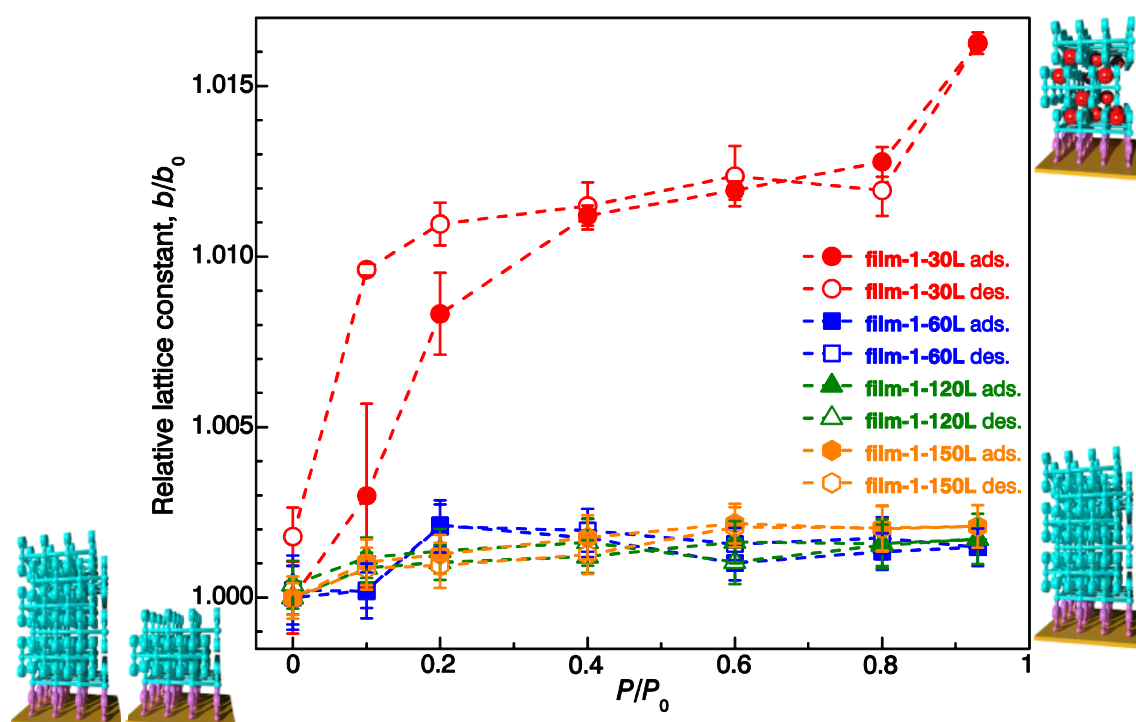


Supplementary Figure 32 | LbL cycle dependency of the 020 diffraction peak ($\lambda = 1.550$

Å). As clearly observed, 020 peak width decreased remarkably with increasing LbL cycles.



Supplementary Figure 33 | LbL cycle dependency of the domain size. D_{OP} and D_{IP} denote the domain sizes along the out-of-plane (i.e., average thickness) and in-plane directions, respectively. These values were calculated from the XRD profiles using Scherrer's equation.



Supplementary Figure 34 | Realization of the dynamical structural transformation induced by thickness downsizing. Relative lattice constant (b/b_0) calculated from 020 and 040 diffraction peak positions in *in situ* out-of-plane XRD measurements on each thin films. The error bars denote $\pm$ s.d. As clearly seen, the structural transformation triggered by EtOH introduction is suppressed remarkably with increasing film thickness, which shows clear thickness downsizing effect of this thin film.

3. Supplementary Notes

References

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